

- Coordinates: $r \equiv \sqrt{\vec{x}^2}$

~D

$$\vec{P} = -i\hbar \vec{\nabla}$$

$$H = \frac{p^2}{2\mu} + V(r)$$

$$H = -\frac{\hbar^2}{2\mu} \nabla^2 + V(r)$$

- $$(\hat{H}\psi)(x) = E\psi = -\frac{\hbar^2}{2\mu} \nabla^2 \psi + V(r)\psi$$

$$i \partial_t \psi = H \psi$$

- $\vec{p}$: momenta

$\vec{L}$: Ang. momenta

E : energy

$$\vec{L} = \vec{x} \times \vec{p} = \vec{x} \times (-i\hbar \vec{\nabla}) = -i\hbar \vec{x} \times \vec{\nabla}$$

- For $\vec{x} = (x_1, x_2, x_3)$

$$L_i = -i\hbar \sum_{jk} \epsilon_{ijk} x_j \frac{\partial}{\partial x_k}$$

$$\epsilon_{ijk} = \begin{cases} +1 & , \text{ if } ijk \text{ even permutation } 123 \\ -1 & , \text{ if } ijk \text{ odd " " } \\ 0 & \text{ otherwise} \end{cases}$$

- To see that $\vec{L}$ is an observable it must commute w/ the Hamiltonian $\hat{H}$:

$$[\hat{L}, \hat{H}] = 0$$

- Consider $[\frac{\partial}{\partial x_i}, x_j] \psi$:

$$\frac{\partial}{\partial x_i} (x; \psi) = x \cdot \left(\frac{\partial}{\partial x_i} \psi \right)$$

$$= \delta_{kj} \psi + x_j \left(\frac{\partial}{\partial x_k} \psi \right) - x_j \left(\frac{\partial}{\partial x_k} \right) \psi$$

$$= \delta_{kj} \psi$$

or

$$\left[\frac{\partial}{\partial x_k}, x_j \right] = \delta_{kj}$$

Thus: $[L_i, x_j] = -i\hbar \left[\sum_{\ell m} \epsilon_{i\ell m} x_\ell \frac{\partial}{\partial x_m}, x_j \right]$

$$= -i\hbar \sum_{\ell m} \epsilon_{i\ell m} x_\ell \left[\frac{\partial}{\partial x_m}, x_j \right]$$

$$= -i\hbar \sum_{\ell m} \epsilon_{i\ell m} x_\ell \delta_{mj}$$

$$= -i\hbar \sum_{\ell} \epsilon_{i\ell j} x_\ell$$

Swap

$$\epsilon_{i\ell j} = -\epsilon_{ij\ell}$$

$$\therefore [L_i, x_j] = +i\hbar \sum_{\ell} \epsilon_{ij\ell} x_\ell$$

• Now, since $[x_m, \frac{\partial}{\partial x_j}] = -\delta_{jm}$, then

$$[L_i, \frac{\partial}{\partial x_j}] = -i\hbar \sum_{\ell k} \epsilon_{i\ell k} \left[x_\ell \frac{\partial}{\partial x_k}, \frac{\partial}{\partial x_j} \right]$$

$$\begin{aligned} \left[x_\ell \frac{\partial}{\partial x_k}, \frac{\partial}{\partial x_j} \right] &= \left(x_\ell \frac{\partial}{\partial x_k} \right) \frac{\partial}{\partial x_j} - \frac{\partial}{\partial x_j} \left(x_\ell \frac{\partial}{\partial x_k} \right) \\ &= x_\ell \left(\frac{\partial^2}{\partial x_k \partial x_j} \right) - \delta_{jl} \frac{\partial}{\partial x_k} - x_\ell \left(\frac{\partial^2}{\partial x_j \partial x_k} \right) \\ &= -\delta_{jl} \frac{\partial}{\partial x_k} \end{aligned}$$

$$\Rightarrow [L_i, \frac{\partial}{\partial x_j}] = -i\hbar \sum_{\ell k} \epsilon_{i\ell k} \left(-\delta_{jl} \frac{\partial}{\partial x_k} \right)$$

$$[L_i, \frac{\partial}{\partial x_j}] = i\hbar \sum_k \epsilon_{ijk} \frac{\partial}{\partial x_k}$$

• Use the above two to show:

$$[L_i, L_j] = i\hbar \sum_k \epsilon_{ijk} L_k$$

SU(2) - Algebra

- In general, if v_j is any vector constructed from $\vec{x}$ and $\vec{v}$ then the following holds
 $[L_i, v_j] = i\hbar \sum_k \epsilon_{ijk} v_k$

Claim: ϵ_{ijk} is zero if two of its entries are equal

Proof: ϵ_{ijk} is totally anti-symmetric and so

$$\epsilon_{ijk} = -\epsilon_{jik}$$

Now, suppose $i = j$, then

$$\epsilon_{iik} = -\epsilon_{iik}$$

$$2\epsilon_{iik} = 0 \Rightarrow \epsilon_{iik} = 0$$

- The above holds for all indices.
- Therefore, for $[L_i, L_i]$ and $i = j$:

$$[L_i, L_i] = i\hbar \sum_k \underbrace{\epsilon_{iik}}_0 L_k = 0$$

$$[L_i, L_i] = 0$$

ex: $[L_1, L_1] = [L_2, L_2] = [L_3, L_3] = 0$

- For different L :

$$[L_1, L_2] = i\hbar \sum_k \epsilon_{12k} L_k$$

k can only be 3 as 1 or 2 yield zero.

$$\underline{[L_1, L_2] = i\hbar L_3}$$

and similarly for other L 's.

* Finally, to show that L_i commute w/ the Hamiltonian.

Hamiltonian for an

- (i) Let V_j be any vector such that
 $[L_i, V_j] = i\hbar \sum_k \epsilon_{ijk} V_k$
- (ii) Then for $|\vec{V}|^2 = \sum_{j=1}^3 V_j V_j$

$$[L_i, |\vec{V}|^2] = [L_i, \sum_{j=1}^3 V_j V_j] = L_i (\sum_j V_j V_j) - (\sum_j V_j V_j) L_i$$

• For Any operators A, B, C :

$$[A, BC] = [A, B]C + B[A, C]$$

proof:

$$[A, BC] = ABC - BCA$$

$$\rightarrow BCA = B(A C + [C, A])$$

$$BCA = BAC + B[C, A]$$

$$= (AB + [B, A])C + B[C, A]$$

$$= ABC + [B, A]C + B[C, A]$$

$$= ABC - [A, B]C - B[A, C]$$

and so

$$[A, BC] = ABC - (ABC - [A, B]C - B[A, C])$$
$$= [A, B]C + B[A, C]$$

Now, we see that

$[L_i, V^2]$ is found for:

$$A = L_i, \quad B = V_j, \quad C = V_j$$

$$[L_i, \sum_j V_j V_j] = \sum_j [L_i, V_j] V_j + \sum_j V_j [L_i, V_j]$$

$\underbrace{\quad}_{i\hbar \sum_k \epsilon_{ijk} L_k}$

$$= \sum_j (i\hbar \sum_k \epsilon_{ijk} V_k) V_j + \sum_j V_j (i\hbar \sum_k \epsilon_{ijk} V_k)$$

$$= i\hbar \sum_{j,k} \epsilon_{ijk} (V_k V_j + V_j V_k)$$

Since ϵ_{ijk} is anti symmetric:

in the second term $\epsilon_{ijk} = -\epsilon_{ikj}$

But we then relabel: $k \rightarrow j$
 $j \rightarrow k$

(Second term):

$$\sum_{k,j} \epsilon_{ijk} V_j V_k = - \sum_{k,j} \epsilon_{ikj} V_k V_j \quad (\text{cancels first term})$$

and so

$$\underline{[L_i, V^2]} = 0$$

(iii) $[L_i, \vec{x}^2] = 0$ so any function
with $r = \sqrt{\vec{x}^2}$ means

$$[L_i, V(r)] = 0$$

$$(iv) [L_i, \nabla^2] = 0$$

Thus for $\hat{H} = -\frac{\hbar^2}{2\mu} \nabla^2 + V(r)$

$$\underline{[L_i, \hat{H}] = 0}$$

i.e. L_i is an observable of the theory.

→ This is the observable associated to the

Rotational symmetry of the Hamiltonian.

Symmetry of theory $\Rightarrow$ (Observable)

* Note what I am calling L_i is:

$$\left. \begin{aligned} L_1 &= L_x \\ L_2 &= L_y \\ L_3 &= L_z \end{aligned} \right\} \quad \left(\begin{array}{l} \text{Notation in} \\ \text{your notes} \end{array} \right)$$

* Important thing to note:

We found

$$[L_i, V^2] = 0$$

for any vector $V_i = V_i(x, \frac{\partial}{\partial x})$ which is
comprised of the operators $\hat{x}$ and $\partial/\partial x$.

Define: $\vec{L}^2 \equiv \sum_i L_i L_i = L_x^2 + L_y^2 + L_z^2$

then

$$[L_i, \vec{L}^2] = 0$$

L_i : Angular momentum in the " i th" direction

$\vec{L}^2$: Total Angular momentum

Claim: $\vec{L}^2$ also commutes with the Hamiltonian
 $[H, \vec{L}^2] = 0$

proof: Recall $[A, BC] = [A, B]C + B[A, C]$

then $[H, \sum_i L_i L_i]$

$A = H, B = L_i, C = L_i$

$$[H, \vec{L}^2] = [H, \sum_i L_i L_i] = \underbrace{[H, L_i] L_i}_{=0} + L_i \underbrace{[H, L_i]}_{=0}$$

Thus

$$[H, \vec{L}^2] = 0$$

So the total angular momentum is conserved and is an observable.

* We can characterize physical states by eigenvalues of:

$$(H, L_z, \vec{L}^2)$$

By convention we choose $L_z = L_3$ as the component of $\vec{L}$ which we use as an eigenvector.

→ To make better sense of why we can only choose one direction:
 Choose L_3 ; then

$$[\hat{H}, \hat{L}_3] = 0$$

Now, since (as shown above)

Also, if we know exactly what L_z is, then we cannot know L_x or L_y

$$[L_1, L_2] = i\hbar L_3$$

$$[L_1, L_3] = -i\hbar L_2$$

$$[L_2, L_3] = i\hbar L_1$$

from the unc.
principle.

L_x and L_y do not commute with L_z and hence do not commute with $\hat{H}$.

Why Polar Coordinates?

Since $[L_z, \vec{x}^2] = [L_z, r] = 0$, the ang. mom. in the z -direction commutes with r , the L_z can act only in the direction of r and not on its length.

- Another way of thinking about this is that $V(r)$ is isotropic and so it should only depend on the orientation.

~D In spherical coordinates: (See notes pg

$$L_1 = L_x = i\hbar \left(\sin(\theta) \frac{\partial}{\partial \theta} + \cot(\theta) \cos(\phi) \frac{\partial}{\partial \phi} \right)$$

$$L_2 = L_y = i\hbar \left(-\cos(\theta) \frac{\partial}{\partial \theta} + \cot(\theta) \sin(\phi) \frac{\partial}{\partial \phi} \right)$$

$$L_3 = L_z = -i\hbar \frac{\partial}{\partial \phi}$$

Also shows us why we choose L_z as it is the simplest to work with!

* A very fun exercise can be done by plugging the above into $\vec{L}^2 = L_x^2 + L_y^2 + L_z^2$ (Pg 68)

or:

$$\vec{L}^2 = -\hbar^2 \left(\frac{1}{\sin(\theta)} \frac{\partial}{\partial \theta} \left(\sin(\theta) \frac{\partial}{\partial \theta} \right) + \frac{1}{\sin^2(\theta)} \frac{\partial^2}{\partial \phi^2} \right)$$

Note that it is independent of r (as it should be)

Quick Aside: If you're still a bit confused about coordinate transformations let's do a quick example for $L_3 = L_z$

$$L_z = i\hbar \left(x_1 \frac{\partial}{\partial x_2} - x_2 \frac{\partial}{\partial x_1} \right)$$

$$x = x_1 = r \sin(\theta) \cos(\phi)$$

$$y = x_2 = r \sin(\theta) \sin(\phi)$$

(Pg 43)

• How do we deal with $\partial/\partial x_i$?

→ Use the Chain Rule

* Recall from Kindergarten:

$$\frac{d}{dx} f(g(x)) = f'(g(x)) \cdot g'(x) \quad \left(\begin{array}{c} \text{chain-} \\ \text{Rule} \end{array} \right)$$

example: $\frac{d}{dx} (\sqrt{x^2+1})$

$$f(x) = \sqrt{x}$$

$$g(x) = x^2 + 1$$

$$\Rightarrow f(g(x)) = \sqrt{x^2 + 1}$$

$$\rightarrow f'(g(x)) \cdot g'(x)$$

$$\frac{1}{2} (x^2 + 1)^{-1/2} \cdot 2x$$

Lets write this in a more useful way:

(i) Let $y = g(x)$

(ii) Then

$$\frac{d}{dx} (f(g(x))) = f'(y) \cdot \frac{dy}{dx} = \frac{df}{dy} \frac{dy}{dx}$$

or $\frac{df}{dx} = \frac{dy}{dx} \frac{df}{dy}$

(iii) I have written the above in a suggestive form. Let us think of $\frac{d}{dx}$ as a Derivative Operator which is acting on some f . Then since

is acting on some f . Then, since f was arbitrary:

$$\frac{d}{dx} = \frac{dy}{dx} \frac{d}{dy}$$

This is a VERY important idea to understand
- It's one of the main tools used in Einstein's theory of Relativity!

(i.v) Now, the above is true for single-variables. When working with more than 1-coordinate the above is written as

$$\frac{d}{dx_i} = \sum_j \frac{dy_j}{dx_i} \frac{d}{dy_j}$$

For spherical coordinates:

$$\left. \begin{aligned} x_1 &= x \\ x_2 &= y \\ x_3 &= z \end{aligned} \right\} \quad \left. \begin{aligned} y_1 &= r \\ y_2 &= \theta \\ y_3 &= \phi \end{aligned} \right\}$$

IMPORTANT: this relation goes both ways meaning, we can write the above as

$$\frac{d}{dy_i} = \sum_j \frac{dx_j}{dy_i} \frac{d}{dx_j}$$

• This is allowed by the beautiful statement that is the Implicit Function Theorem.

→ Back to $L_z = i\hbar (x_1 \frac{\partial}{\partial x_2} - x_2 \frac{\partial}{\partial x_1})$
Let us start with

$$\frac{\partial}{\partial \phi} = \frac{\partial}{\partial x_3} = \sum_j \frac{\partial x_j}{\partial \phi} \frac{\partial}{\partial x_j} = \frac{\partial x_1}{\partial \phi} \frac{\partial}{\partial x_1} + \frac{\partial x_2}{\partial \phi} \frac{\partial}{\partial x_2} + \frac{\partial x_3}{\partial \phi} \frac{\partial}{\partial x_3}$$

$$= \frac{\partial x}{\partial \phi} \frac{\partial}{\partial x} + \frac{\partial y}{\partial \phi} \frac{\partial}{\partial y} + \frac{\partial z}{\partial \phi} \frac{\partial}{\partial z}$$

we know

$$x = r \sin(\theta) \cos(\phi), y = r \sin(\theta) \sin(\phi), z = r \cos(\theta)$$

$$\frac{\partial x}{\partial \phi} = -r \sin(\theta) \sin(\phi) = -y = -x_2$$

$$\frac{\partial y}{\partial \phi} = r \sin(\theta) \cos(\phi) = x = x_1$$

$$\frac{\partial z}{\partial \phi} = 0$$

Plugging this in:

$$\frac{\partial}{\partial \phi} = -x_2 \frac{\partial}{\partial x_1} + x_1 \frac{\partial}{\partial x_2} = - \left(x_1 \frac{\partial}{\partial x_2} - x_2 \frac{\partial}{\partial x_1} \right) = - \frac{L_z}{i\hbar}$$

or

$$L_z = -i\hbar \frac{\partial}{\partial \phi}$$

as claimed.

L_x, L_y are found similarly.

* So far, we've found that a single Particle Subject to an isotropic potential $V(r)$ is, in QM, characterized by

(i) E : energy ($\hat{H}$: Hamiltonian)

(ii) $\hat{L}_z$: ang. mom. orientation

(iii) $\hat{L}^2$: total ang. momenta.

• We have the tools to express $\hat{L}_z$ and $\hat{L}^2$ in terms of coordinates (x, y, z) or (r, θ, ϕ) but we have yet to find out the eigenvalues associated to these operators.

These eigenvalues are what we are able to "measure" and hence are of great importance.

Let us now work towards that:

• As you can verify:

$$\vec{\nabla}^2 = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial}{\partial r} \right) - \frac{\vec{L}^2}{\hbar^2 r^2} \quad (\text{Pg 45})$$

So we can plug this into the (time-ind.) Schrödinger Eq:

$$(\hat{H}\psi)(x) = E\psi(x) = -\frac{\hbar^2}{2\mu} \underbrace{\vec{\nabla}^2}_{\text{Replace}} \psi(x) + V(r)\psi(x)$$

$$E\psi(x) = -\frac{\hbar^2}{2\mu r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial \psi(x)}{\partial r} \right) + \frac{(\vec{L}^2 \psi)(x)}{2\mu r^2} + V(r)\psi(x)$$

• As is, there are some assumptions we can make to make the above problem simpler. This is done, as with the 1D Harmonic Oscillator, by looking at the operator alg.

→ This means we find a complete set of ops which (here) define a Cartan algebra: i.e. we find raising & lowering operators.

(i) Suppose that we label the eigenvalue of L_z by m :

$$L_z |\psi\rangle = \hbar m |\psi\rangle$$

(ii) Recall for the 1D Harmonic Osc.

$$H = p^2/2m + m\omega^2 x^2/2$$

if we split this up Classically

$$a = \left(\frac{m\omega}{2\hbar}\right)^{1/2} \left(X + \frac{iP}{m\omega}\right) \quad a^\dagger = \left(\frac{m\omega}{2\hbar}\right)^{1/2} \left(X - \frac{iP}{m\omega}\right)$$

$$aa^\dagger = \left(\frac{m\omega}{2\hbar}\right) \left(X^2 + \frac{P^2}{m^2\omega^2}\right)$$

$$= (m\omega/2\hbar) X^2 + \frac{P^2}{2\hbar\omega m}$$

$$\hbar\omega aa^\dagger = \frac{P^2}{2m} + \frac{m\omega^2 X^2}{2} = "H" \text{ (Classical)}$$

• What is the point to this? we see that we have "factored" the Hamiltonian into two new operators:

a : lowering op

$a^\dagger$: Raising op

Quantum:

$$H = (aa^\dagger + \frac{1}{2})\hbar\omega \text{ (Quantum)}$$

• How do we transfer this knowledge to $\vec{L}^2$? We look at the Hamiltonian

$$\hat{H} = \underbrace{-\frac{\hbar^2}{2mr^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial}{\partial r}\right) + V(r)}_{\text{Radial}} + \underbrace{\frac{1}{2mr^2} \vec{L}^2}_{\text{Angular}}$$

The radial part is of no importance to us since $\vec{L}^2$ and L_z only depend on (θ, ϕ) . So, we can look at "factoring" the angular part as we did for the 1D SHO.

$$L^2 = L_x^2 + L_y^2 + L_z^2$$

Because L_z and $\vec{L}^2$ have distinct eigenvalues we would expect L_z to be

expressible via L_+ and L_- only, and well, it is! :

$$\begin{aligned} L_+ &\equiv L_x + iL_y \\ L_- &\equiv L_x - iL_y \end{aligned}$$

Claim: $\vec{L}^2 = L_+ L_- + L_- L_+ + L_z^2 - \hbar L_z$

Proof:

$$\begin{aligned} &= L_x^2 + L_y^2 + i(L_y L_x - L_x L_y) + L_z^2 - \hbar L_z \\ &= L_x^2 + L_y^2 - i[L_x, L_y] + L_z^2 - \hbar L_z \\ &= L_x^2 + L_y^2 - i(i\hbar L_z) + L_z^2 - \hbar L_z \\ &= L_x^2 + L_y^2 + \hbar L_z + L_z^2 - \hbar L_z \\ &= \vec{L}^2 \end{aligned}$$

Now, in order for these to be raising & lowering ops they must satisfy

$$\begin{aligned} \text{(a)} \quad L_z (L_+ |m\rangle) &= (m+1) (L_+ |m\rangle) \\ \text{(b)} \quad L_z (L_- |m\rangle) &= (m-1) (L_- |m\rangle) \end{aligned}$$

In order to prove the above we must first show the following:

Claim: $[L_z, L_+] = \hbar L_+$

Proof: $[L_z, (L_x + iL_y)] = [L_z, L_x] + i[L_z, L_y]$
 $= i\hbar L_y + \hbar L_x = \hbar(L_x + iL_y)$
 $= \hbar L_+$

Similarly $[L_z, L_-] = -\hbar L_-$ (show this)

Use these to prove (a) and (b)

(a) proof:

$$\begin{aligned} (L_z L_+ |m\rangle) &= (L_+ L_z + [L_z, L_+]) |m\rangle \\ &= (L_+ L_z + \hbar L_+) |m\rangle \\ &= L_+ (L_z |m\rangle) + \hbar L_+ |m\rangle \end{aligned}$$

$$L_z(L+|m\rangle) = \hbar(m+1)(L+|m\rangle)$$

as required

$$\hbar m/m \rightarrow$$

Recall we assumed

→ Do this for (b) as well.

$$L_z|4\rangle = \hbar m|4\rangle$$

• m : eigenvalues of L_z

• $|m\rangle$: eigenvectors associated to L_z
 $L_z|m\rangle = \hbar m|m\rangle$

• L_+ : Raising operator associated to $|m\rangle$

• Since L_+ raises the value of m by 1 it takes us to the next level state:

$$\underbrace{C_{m+1}}_{\text{some constant}} |m+1\rangle = L_+ |m\rangle \quad (\text{notation})$$

• L_- : Same but lowers

$$\underbrace{C_{m-1}}_{\text{some constant}} |m-1\rangle = L_- |m\rangle \quad (\text{notation})$$

* Next step: find Eigenvalues for $\vec{L}^2$

→ Assume $\vec{L}^2 |m, \alpha\rangle = \hbar^2 \alpha |m, \alpha\rangle$

(we are simply adding a label to $|m\rangle$
 but they are the same states as above.)

• Recall $[L_i, \vec{L}^2] = 0$, from this we easily conclude

$$\begin{aligned}[L_x, L^2] &= 0 \\ [L_y, L^2] &= 0 \\ [L_+, L^2] &= 0 \\ [L_-, L^2] &= 0\end{aligned}$$

and so

$$\begin{aligned}\vec{L}^2 L_+ |m, \alpha\rangle &= L_+ \vec{L}^2 |m, \alpha\rangle \\ \vec{L}^2 L_- |m, \alpha\rangle &= L_- \vec{L}^2 |m, \alpha\rangle\end{aligned}$$

- L_+, L_- only change $\hat{L}_z$ eigen values but not $\vec{L}^2$.
- $\vec{L}^2 (L_{\pm} |m, \alpha\rangle) = \hbar^2 \alpha (L_{\pm} |m, \alpha\rangle)$
- from here (pg 50 of the notes gives a vague explanation, here we are precise) we will find limits for (m, α)

upper limit: There is an upper limit on the eigenvalue α of $\vec{L}^2$.

- Recall any physical operator is positive i.e.

$$\langle m, \alpha | \hat{A} | m, \alpha \rangle \geq 0$$

Let

$$\hat{A} = \vec{L}^2 - \hat{L}_z^2$$

then

$$\begin{aligned}\langle m, \alpha | \vec{L}^2 - \hat{L}_z^2 | m, \alpha \rangle &= \langle m, \alpha | \vec{L}^2 | m, \alpha \rangle - \langle m, \alpha | \hat{L}_z^2 | m, \alpha \rangle \\ &= \hbar^2 \alpha - \hbar^2 m^2 \geq 0\end{aligned}$$

and so

$$\boxed{\alpha \geq m^2}$$

- Lets find this conceptually:

$$\vec{L}^2 = L_x^2 + L_y^2 + L_z^2$$

now suppose we know the state has ang. momenta

$$\vec{L}^2 = \alpha$$

$$\text{with } L_x = \hbar m \quad L_y = \hbar \sqrt{\alpha - m^2} \quad L_z = \hbar m$$

with $L_x = 0$, $L_y = 0$
then

$$\alpha = \gamma^2 + \delta^2 + m^2$$

then it is clear that if $L_x = L_y = 0$
then m will have a max of
 $\alpha = m^2$

(See pg's 52 & 54 for figures)
For non-zero L_x, L_y , m^2 is less than
 α .

Since we have a max value for m
 $m = m_{\max}$
and a minimum
 $m = m_{\min}$

(This is called
the highest
weight state)

there is the "max" state $|\alpha, m_{\max}\rangle$
where we can no longer go "up"

$$L_+ |\alpha, m_{\max}\rangle = 0$$

Since: $L \cdot L_+ = \vec{L}^2 - L_z^2 - \hbar L_z$ (see above)

$$L_- L_+ |\alpha, m_{\max}\rangle = \vec{L}^2 |\alpha, m_{\max}\rangle - L_z^2 |\alpha, m_{\max}\rangle - \hbar L_z |\alpha, m_{\max}\rangle = 0$$

$$\hbar^2 \alpha |\alpha, m_{\max}\rangle - \hbar^2 m_{\max}^2 |\alpha, m_{\max}\rangle - \hbar^2 m_{\max} |\alpha, m_{\max}\rangle = 0$$

$$\alpha - m_{\max}^2 - m_{\max} = 0$$

or

$$\alpha = m_{\max} (m_{\max} + 1)$$

* Likewise, for the minimum

$$L_- |\alpha, m_{\min}\rangle = 0$$

and using $\vec{L}^2 = \hat{L}_x^2 + \hat{L}_y^2 + \hat{L}_z^2$

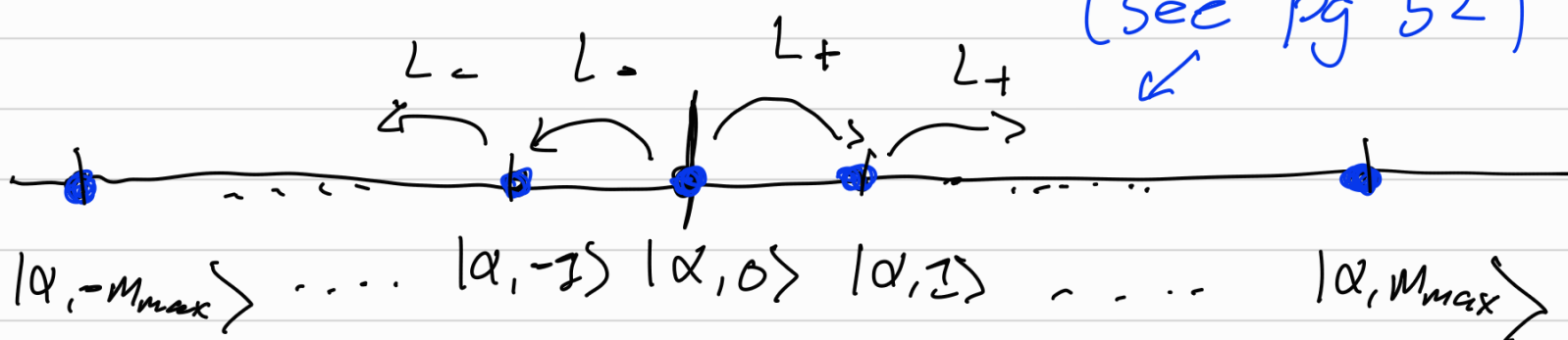
$$\alpha = m_{\min}(m_{\min} - 1)$$

for which we have

$$m_{\min} = -m_{\max}$$

as expected from

$$\alpha \geq m^2 \quad (\text{see pg 52})$$



Labels: * $l \equiv m_{\max}$ *

$$\Rightarrow \alpha = l(l+1)$$

Since we assumed $\vec{L}^2 |\alpha, m\rangle = \hbar^2 \alpha |\alpha, m\rangle$ we have

$$\vec{L}^2 |l, m\rangle = l(l+1)\hbar^2 |l, m\rangle$$

→ Notice the change of label from $|\alpha, m\rangle$ to $|l, m\rangle$ since $\alpha = l(l+1)$ it would be a bit annoying to write $|l(l+1), m\rangle$ so we write $|l, m\rangle$ instead.

We therefore conclude

$$\hat{L} |l, m\rangle = \hbar \sqrt{l(l+1)} |l, m\rangle$$

there are a total of $(2l+1)$ - states

ex: $l=0$!

$$|0, 0\rangle$$

1

$|l, m\rangle$

$l=1$:

$$|1, -1\rangle \xleftarrow{L_-} |1, 0\rangle \xrightarrow{L_+} |1, 1\rangle$$

3

* See pg 53 for summary

Separation of Variables:

Notation

$$\left(\begin{array}{l} \partial_x \equiv \frac{\partial}{\partial x} \\ \partial_t \equiv \frac{\partial}{\partial t} \end{array} \right)$$

$$-\frac{\hbar^2}{2\mu} \nabla^2 \tilde{\psi} + V(r) \tilde{\psi} = i\hbar \partial_t \tilde{\psi}$$

$$\nabla^2 = \partial_x^2 + \partial_y^2 + \partial_z^2$$

Has a solution

$$\tilde{\psi}(\vec{r}, t) = \psi(\vec{r}) e^{-iEt/\hbar}$$

(See chapter
6 of textbook
I sent)

$$-\frac{\hbar^2}{2\mu} \nabla^2 \psi + V(r) \psi = E \psi$$

now, suppose $V(r) = V(x) + V(y) + V(z)$

then

$$-\frac{\hbar^2}{2\mu} (\partial_x^2 + \partial_y^2 + \partial_z^2) \psi + (V(x) + V(y) + V(z)) \psi = E \psi$$

combining terms

$$\left[\left(-\frac{\hbar^2}{2m} \partial_x^2 + V(x) \right) + \left(-\frac{\hbar^2}{2m} \partial_y^2 + V(y) \right) + \left(-\frac{\hbar^2}{2m} \partial_z^2 + V(z) \right) \right] \psi = E \psi$$

which is of the form $\hat{H} \psi = E \psi$, note that the Hamiltonian has completely split into the sum of x, y, z terms

Define: $\hat{H}_x \equiv -\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} + V(x)$

and similarly for $\hat{H}_y, \hat{H}_z$ so that $\hat{H} = \hat{H}_x + \hat{H}_y + \hat{H}_z$

• When the differential operator splits as above we can use the method of Separation of variables:

$$\psi(x, y, z) = X(x) Y(y) Z(z)$$

This means ψ factors into the product of coordinate functions (as above)

$$(\hat{H}_x + \hat{H}_y + \hat{H}_z) (X(x) Y(y) Z(z)) = E X(x) Y(y) Z(z)$$

$$(\hat{H}_x X(x)) Y(y) Z(z) + X(x) (\hat{H}_y Y(y)) Z(z) + X(x) Y(y) (\hat{H}_z Z(z)) = E X(x) Y(y) Z(z)$$

• Divide both sides by $X(x) Y(y) Z(z)$
Left with

$$\frac{1}{X(x)} \hat{H}_x X(x) + \frac{1}{Y(y)} \hat{H}_y Y(y) + \frac{1}{Z(z)} \hat{H}_z Z(z) = E$$

Let $E = E_x + E_y + E_z$, then the above separates into three similar Equations:

$$\frac{1}{X(x)} \left(-\frac{\hbar^2}{2m} \frac{d^2 X(x)}{dx^2} + V(x) X(x) \right) = E_x$$

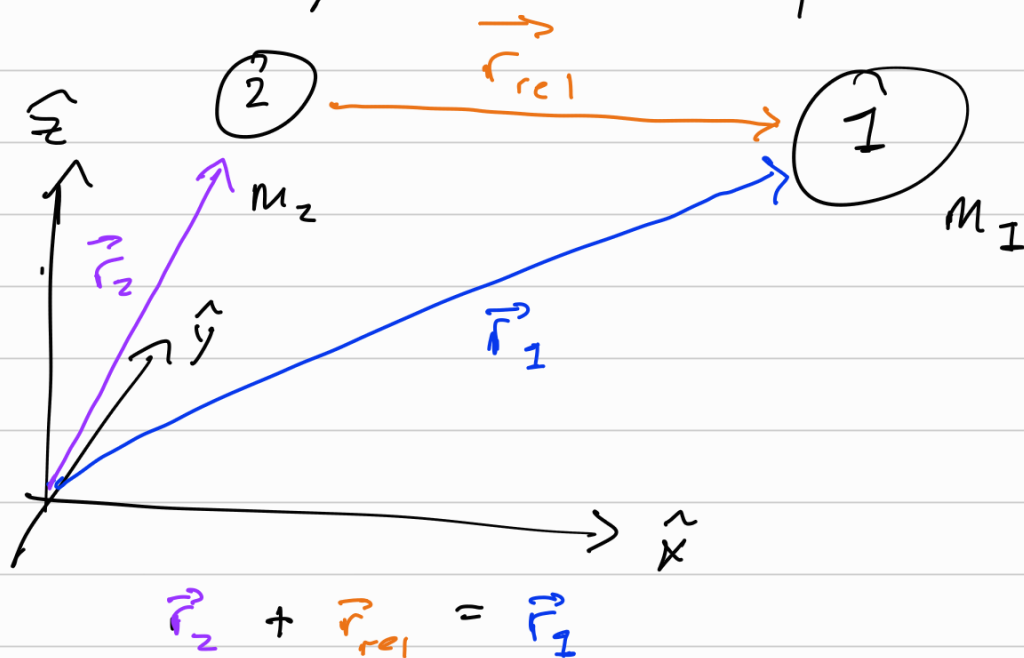
$$\text{or} \quad \frac{-\hbar^2}{2\mu} \frac{d^2 \bar{\chi}(x)}{dx^2} + V(x) \bar{\chi}(x) = E_x \bar{\chi}(x)$$

and similarly for $\bar{\Psi}(y)$ & $\bar{Z}(z)$.

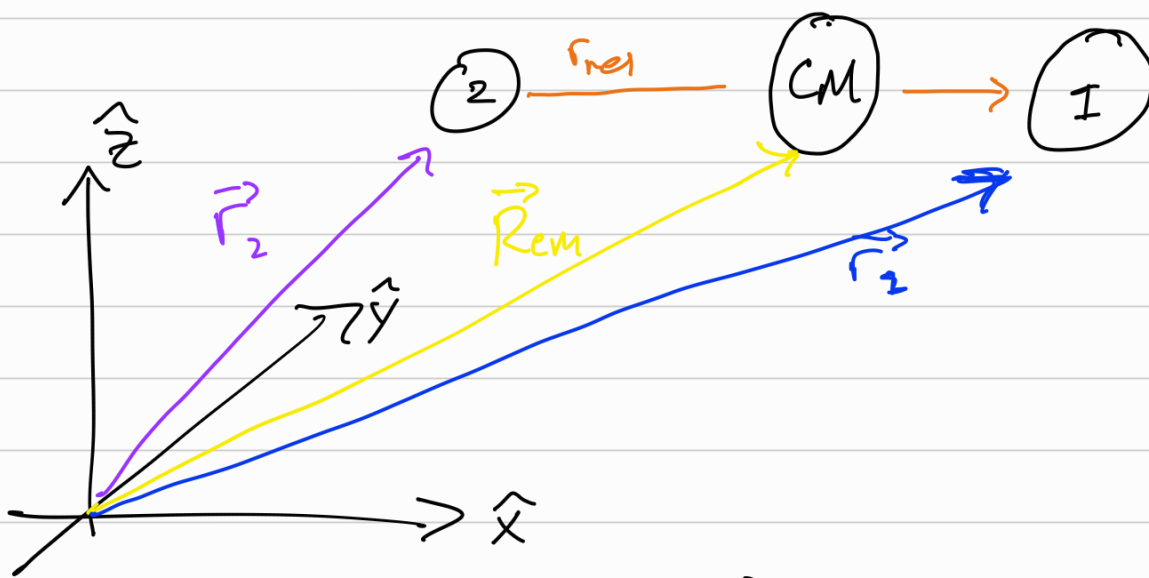
- We can then solve this for a given $V(x)$.
- Can go over method of solving if need be.

Two body system:

- Classically: Take two particles



Center of Mass:



- Momenta: $\vec{P}_{cm} = \vec{P}_1 + \vec{P}_2$
 $\vec{P}_1 = m_1 \vec{v}_1$
 $\vec{P}_2 = m_2 \vec{v}_2$
 $\vec{P}_{cm} = (m_1 + m_2) \vec{v}_{cm}$

$\therefore$

$$(m_1 + m_2) \vec{v}_{cm} = m_1 \vec{v}_1 + m_2 \vec{v}_2$$

which gives us the position of the C.O.M.:

$$\vec{R}_{cm} = \frac{1}{(m_1 + m_2)} (m_1 \vec{r}_1 + m_2 \vec{r}_2)$$

- The relative momenta:

$$\vec{v}_{rel} = \vec{v}_1 - \vec{v}_2$$

$$\hat{p}_{rel} = \mu \left(\frac{\hat{p}_1}{m_1} - \frac{\hat{p}_2}{m_2} \right) \quad \mu = \frac{m_1 m_2}{m_1 + m_2}$$

$$\frac{\hat{p}_{rel}}{\mu} = \frac{\hat{p}_1}{m_1} - \frac{\hat{p}_2}{m_2} \quad \mu \equiv m_1 + m_2$$

Hamiltonian:

$$\hat{H} = \frac{\hat{p}_1^2}{2m_1} + \frac{\hat{p}_2^2}{2m_2} + V(\vec{r}_1 - \vec{r}_2)$$

replace p_1 and p_2 by $\vec{p}_{cm}$ and $\vec{p}_{rel}$:

$$\hat{H} = \frac{\hat{p}_{cm}^2}{2M} + \frac{\hat{p}_{rel}^2}{2\mu} + V(r_{rel})$$

- Note, as the example above, the Hamiltonian splits into two segments:

$$\hat{H} = \hat{H}_{cm} + \hat{H}_{rel}$$

$$\hat{H}_{cm} = \hat{p}_{cm}^2 / 2M$$

$$\hat{H}_{rel} = \hat{p}_{rel}^2 / 2\mu + V(r_{rel})$$

This means we can split $\psi(r)$ into

$$\psi(r) = \psi_{cm}(R_{cm}) \psi_{rel}(r_{rel})$$

and use separation of variables on

$$\hat{H}\psi = E\psi = (\hat{H}_{cm} + \hat{H}_{rel})\psi$$

$$\text{for } E_{cm} + E_{rel} = E$$

$$\left[-\frac{\hbar^2 \vec{\nabla}_{cm}^2}{2M} (\psi_{cm}(R_{cm})) \right] \psi_{rel}(r_{rel}) + \psi_{cm}(R_{cm}) \left[-\frac{\hbar^2 \vec{\nabla}_{rel}^2}{2\mu} \psi_{rel} + V(r_{rel}) \psi_{rel} \right] = E \psi_{cm} \psi_{rel}$$

Dividing through by $\psi_{cm} \psi_{rel}$:

$$\frac{1}{\psi_{cm}} \left(-\frac{\hbar^2}{2M} \vec{\nabla}_{cm}^2 \psi_{cm} \right) + \left[-\frac{\hbar^2}{2\mu} \vec{\nabla}_{rel}^2 \psi_{rel} + V(r_{rel}) \psi_{rel} \right] = E$$

and so we are solving the two systems

$$-\frac{\hbar^2}{2M} \nabla_{cm}^2 \psi_{cm}(R_{cm}) = E_{cm} \psi_{cm}(R_{cm})$$

$$-\frac{\hbar^2}{2\mu} \nabla_{rel}^2 \psi_{rel}(r_{rel}) + V(r_{rel}) \psi_{rel}(r_{rel}) = E_{rel} \psi_{rel}(r_{rel})$$

(See pages 72-73)

As $r \rightarrow 0$:

$$\psi(x) \rightarrow r^l \psi(\theta, \phi)$$

$\Rightarrow$

$$\vec{L}^2 \psi = \hbar^2 \frac{\partial}{\partial r} \left(r^2 \frac{\partial \psi}{\partial r} \right) + 2\mu r^2 [E - V(r)] \psi$$

- $V(r)$ must be less singular than $1/r^2$, then second term vanishes.

$$\hbar^2 \frac{\partial}{\partial r} (r^2 \nabla^2 r^{l-1} \psi(\theta, \phi))$$

$$= \hbar^2 \psi(\theta, \phi) l(l+1) r^l$$

$$= \hbar^2 l(l+1) \psi$$

$$\boxed{L^2 \psi \rightarrow \hbar^2 l(l+1) \psi}$$

• True for all r by translational invariance

$$\psi(x) = R(r) Y(\theta, \phi)$$

where

$$R(r) \propto r^l, \quad r \rightarrow 0$$

and

$$\boxed{L^2 Y = \hbar^2 l(l+1) Y}$$

likewise,

$$\boxed{L_z Y = \hbar m Y}$$

$$|\psi\rangle = \sum_{l,m} c_{lm} |l,m\rangle$$

$$\langle \vec{r} | \psi \rangle = \psi(r, \theta, \phi) = \sum_{l,m} c_{lm} \underbrace{\langle \vec{r} | l, m \rangle}_{R_l(r) Y_{lm}(\theta, \phi)}$$

$$\langle \vec{r} | \psi \rangle = \sum_{l,m} c_{lm} R_l(r) Y_{lm}(\theta, \phi)$$

Thus,

$$\hat{L}_z |\psi\rangle = \hbar m |\psi\rangle$$

• Look at single solution:

$$\langle \vec{r} | \psi \rangle = R(r) Y(\theta, \phi)$$

$$\langle \vec{r} | \hat{L}_z | \psi \rangle = \hbar m \langle \vec{r} | \psi \rangle$$

$$-i\hbar \frac{\partial}{\partial \phi} (R(r) Y(\theta, \phi)) = \hbar m (R(r) Y(\theta, \phi))$$

or

$$\boxed{-i \frac{\partial Y}{\partial \phi} = m Y(\theta, \phi)}$$

* Sep of variables:

$$Y(\theta, \phi) = \Theta(\theta) \Phi(\phi)$$

$$\therefore -i \frac{\partial \Phi}{\partial \phi} = m \Phi \Rightarrow \boxed{\Phi = A e^{im\phi}}$$

• Boundary Condition: $\Phi(0) = \Phi(2\pi)$

$$1 = e^{2\pi i m}$$

$$\Rightarrow \boxed{m \in \mathbb{Z}}$$

$$\boxed{\Psi(\theta, \phi) = A e^{im\phi} \Theta(\theta)}$$

Radial part of Eq:

$$E R(r) = -\frac{\hbar^2}{2\mu r^2} \frac{d}{dr} \left(r^2 \frac{dR}{dr} \right) + \frac{\hbar^2 l(l+1)}{2\mu r^2} R(r) + V(r) R(r)$$

• Need $\int d^3x |\Psi|^2 \xrightarrow{r \rightarrow \infty} 0$

$$\left(\int d\theta d\phi \sin(\theta) \Psi^2 \right) \left(\int_0^\infty |R(r)|^2 r^2 dr \right) \rightarrow$$

Thus

$$\int_0^\infty dr |R(r)|^2 r^2 < \infty$$

with $V(r) \xrightarrow{r \rightarrow \infty} 0$

• Solve $R(r)$ via looking at asymptotics of how $R(r)$ behaves at $r \rightarrow 0$ & $r \rightarrow \infty$

• The Θ part comes from

$$\vec{L}^2 |l, m\rangle = \hbar^2 l(l+1) |l, m\rangle$$

$$\langle \vec{r} | \vec{L}^2 |l, m\rangle = \hbar^2 l(l+1) \langle \vec{r} |l, m\rangle$$

$$(\quad) (R(r) e^{im\phi} \Theta(\theta)) = \hbar^2 l(l+1) R(r) e^{im\phi} \Theta(\theta)$$

$$\ast \vec{L}^2 (e^{im\phi} \Theta(\theta)) = \hbar^2 l(l+1) e^{im\phi} \Theta(\theta)$$

• Can use

$$L_+ |l, l\rangle = 0$$

$$\langle \vec{r} | L_+ |l, l\rangle = L_+ (R(r) Y_l^l(\theta, \phi)) = 0$$

$$L_t = L_x + i h_y$$